
MSML 605

MLOps - ML Project lifeCycle

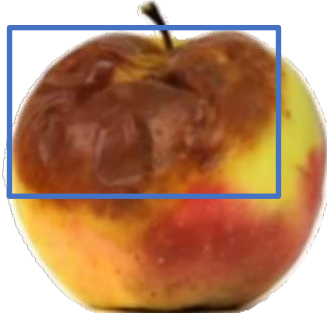
Introduction

- We want to build a ML system for a production environment.
 - Some examples of using ML models:
 - Automotive industry: detect defects like scratches, dents, misalignments etc. on car body panels.
 - Electronics manufacturing: inspect printed circuit boards (PCBs) for defects like, missing components, soldering defects, damaged traces etc.
 - Food processing: inspect food products for defects like, discoloration, bruises, or foreign objects.
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Introduction

- Some examples of using ML models:
 - Textile manufacturing - inspect fabrics for defects such as tears, stains, or weaving irregularities.
 - Pharmaceutical industry - inspect tablets, capsules or vials for defects like cracks, chips, or impurities.
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Training



Training



Vision
Device

Input



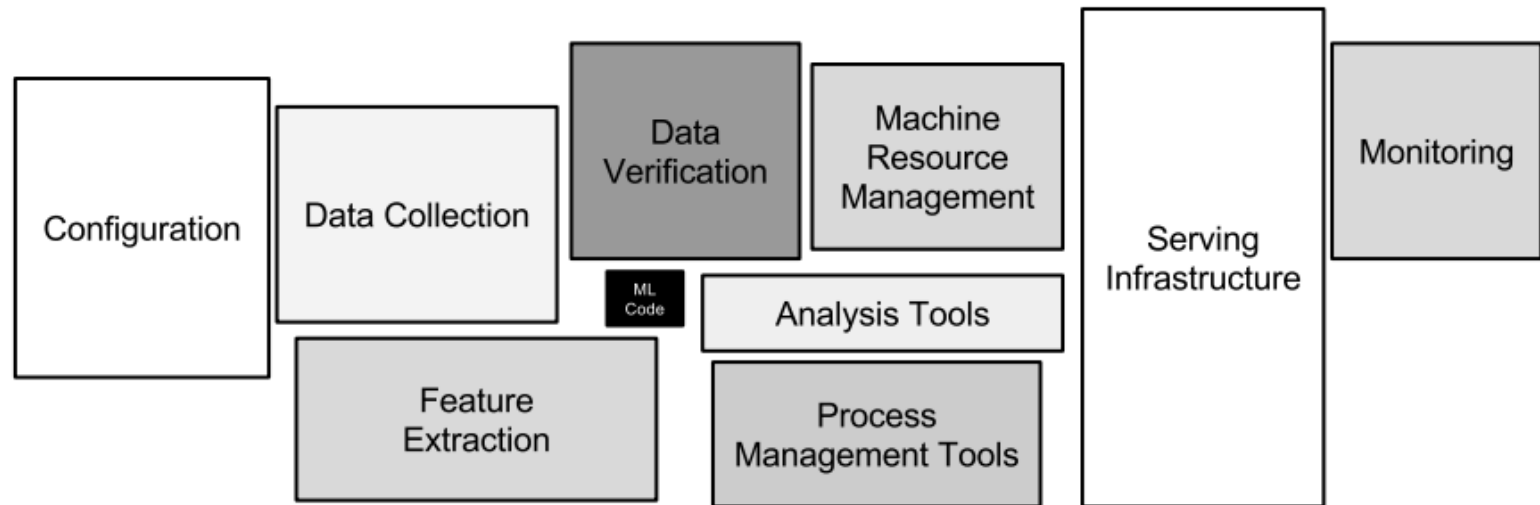
Prediction



Test (Data drift)

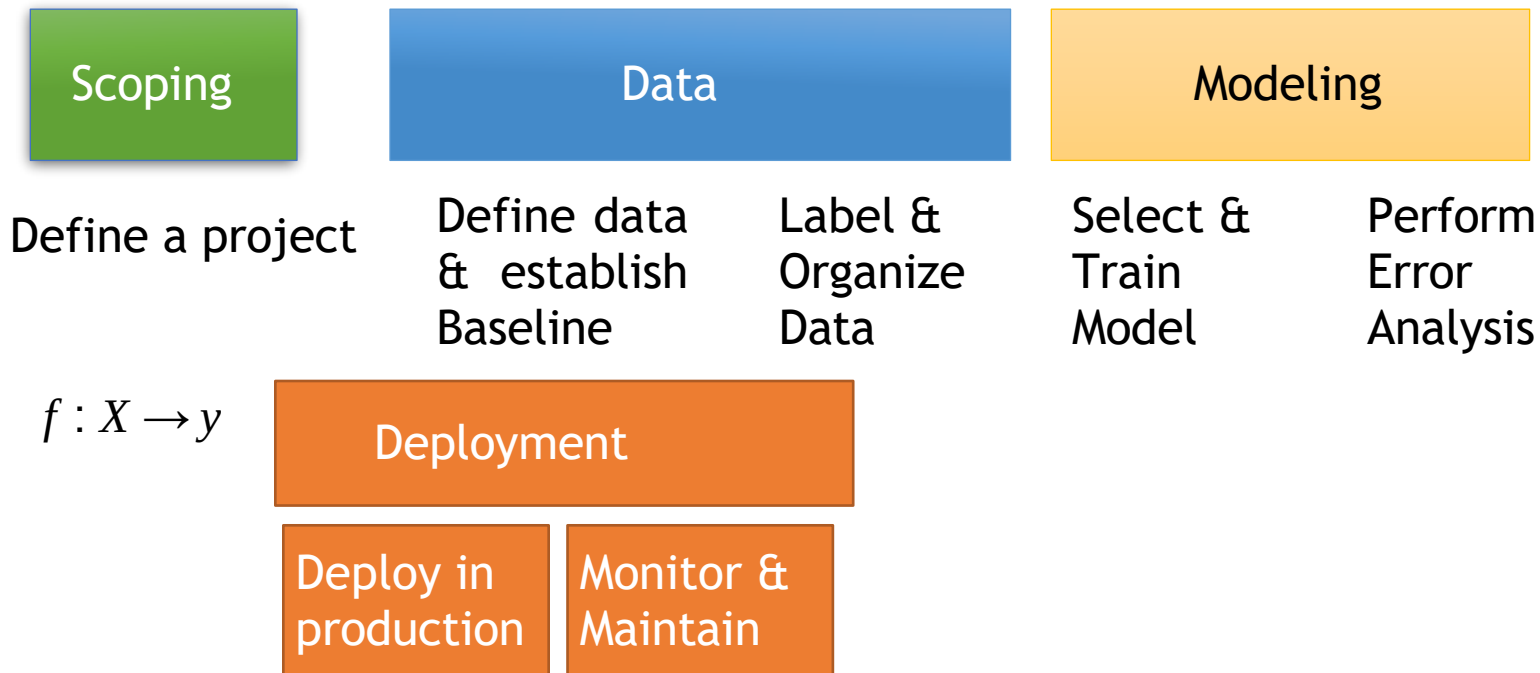


ML System



Hidden Technical Debt in Machine Learning Systems by Sculley et al., NIPS 2015

ML Project: lifecycle



Case Study: Financial Company

- ABC financial company offers financial products & services.
 - data generated from customer transactions, market trends, and economic indicators.
 - Leverage ML to enhance various aspects of their operations, including risk assessment, fraud detection, customer segmentation, and investment strategies.
 - Metrics: depends on the problem
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Case Study: Metrics

- ABC financial company offers financial products & services.
 - Risk Assessment:
 - Accuracy: Proportion of correctly predicted risky instances out of the total instances.
 - Precision: Ratio of the true positive predictions to the total positive predictions made by the model.
 - Recall: Ratio of true positive predictions to the total actual positive instances in the dataset.
 - F1 Score: Harmonic mean of precision and recall.
 - Area under the ROC Curve (AUC-ROC): Ability of the model to distinguish between different classes across various thresholds.
 - Confusion Matrix: A table summarizing the performance of a classification model, showing true positives, true negatives, false positives, and false negatives.

Case Study: Metrics

- ABC financial company offers financial products & services.
 - Fraud detection:
 - False positive rate: The proportion of non-fraudulent transactions incorrectly classified as fraudulent by the model.
 - False Negative Rate: The proportion of fraudulent transactions incorrectly classified as non-fraudulent by the model.
 - Precision, recall & F1 score
 - Detection rate: percentage of fraudulent transactions correctly identified by the model.
 - Transaction Volume: total number of transactions flagged as potentially fraudulent by the model.
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Case Study: Metrics

- ABC financial company offers financial products & services.
 - Customer Segmentation:
 - Silhouette Score: A measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation).
 - Homogeneity Score: A metric that measures the extent to which clusters contain only members of a single class.
 - Completeness score: measures the extent to which all members of a given class are assigned to the same cluster.
 - Rand Index: a measure of the similarity between two data clusterings.
 - Cluster purity: Proportion of data points that belong to the predominant class in that cluster.
 - Inter-cluster distance: Average distance between cluster centroids.

Case Study: Metrics

- ABC financial company offers financial products & services.
 - Investment Strategies:
 - Portfolio Return: total return generated by the investment portfolio over a specific period.
 - Risk-Adjusted Return: return on investment adjusted for the level of risk taken.
 - Sharpe Ratio: Measure of risk-adjusted return, calculated as the portfolio return minus the risk free rate divided by the portfolio's standard deviation.
 - Maximum Drawdown: Maximum loss from a peak to a trough of a portfolio before a new peak is attained.
 - Beta: Measure of a portfolio sensitivity to market movements, indicating its volatility relative to the market as a whole.
 - Alpha: Measure of the excess return of a portfolio compared to the return predicted by its beta
 - Portfolio turnover: rate at which assets in the portfolio are bought and sold, indicating the level of trading activity.

Case Study: ML

- ABC financial company offers financial products & services.
 - Model training
 - Hyperparameter tuning
 - Hold the code fixed and optimize the data and the hyperparameters.
 - After error analysis, deploy
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Case Study: Deployment

- ABC financial company offers financial products & services.
 - Containerization & packaging
 - Version Control
 - Continuous Integration/Continuous Deployment
 - Deployment strategies like blue-green deployments or canary releases are employed to minimize downtime and risk
 - Monitoring tools are integrated into the deployment pipeline to track the performance and health of deployed models in real-time.
 - A/B testing frameworks are utilized to compare the performance of new model versions against existing ones in real-world scenarios.

Case Study: Concept Drift

- ABC financial company offers financial products & services.
 - Statistical properties of the target variable change over time - relationship between features and the target variable.
 - For example, a model trained to predict credit risk based on historical data may encounter concept drift if lending criteria or consumer credit habits change over time.

Case Study: Concept Drift Mitigation

- ABC financial company offers financial products & services.
 - Continuous Monitoring: Sudden drops in performance or deviations from historical patterns may indicate concept drift.
 - Retraining Strategies: Incremental learning techniques can be used to update the model without retraining from scratch.
 - Ensemble Methods: combine predictions from multiple models trained on different time windows or subsets of data.

Case Study: Data Drift

- ABC financial company offers financial products & services.
 - statistical properties of the input data (features) used for prediction change over time
 - For example, shifts in customer preferences or the introduction of new financial products may lead to changes in the distribution of input features.
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Case Study: Data Drift - Mitigation

- ABC financial company offers financial products & services.
 - statistical properties of input features (e.g., mean, variance, distribution) over time to detect shifts indicative of data drift.
 - Robust feature transformations and normalization methods
 - preprocessing pipelines that handle missing values, outliers, and other data quality issues in real-time.
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Case Study: Software Engineering Issues

- ABC financial company offers financial products & services.
 - Realtime or batch?
 - Cloud vs. edge / browser
 - Compute Resources (CPU / GPU / TPU , memory)
 - Latency & throughput
 - Logging
 - Security & privacy
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Case Study: Monitoring

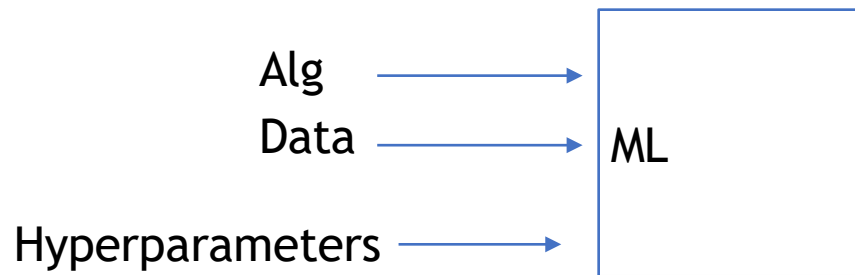
- ABC financial company offers financial products & services.
 - Input metrics
 - Data quality
 - Data volume
 - Feature engineering
 - Regulatory changes
 - Output metrics
 - Model accuracy
 - Prediction latency
 - Risk assessment
 - Fraud detection rate
 - Customer segmentation quality
 - Investment performance
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Case Study: Monitoring

- ABC financial company offers financial products & services.
 - Deployment
 - New data
 - Performance analysis
 - Repeat
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Modeling

- Model Centric vs. Data Centric modeling
- Challenges
 - ML systems - Model (iterative)& Data
 - Flexibility of data - customized data to the problem



Modeling

- Model should do well on the training set
 - Does well on dev/test set.
 - Well on real world metrics.
 - It is not enough to do well on the unknown test set.
 - Informational or transactional queries
 - Learning about technology, a place to visit, player statistics, political developments in a region etc.
 - UMD, Reddit etc - navigational queries
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Performances on key parts of dataset

- ML for loan approval
 - Do not discriminate based on location, language, gender, ethnicity etc.
 - Airline recommendations
 - Fairly treat all partners or categories.
 - Internet content delivery
 - Skewed data distribution (fraud, medical diagnosis (lot of negative examples))
 - Example chicken behavior.
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Baseline - human level performance

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Baseline - Structured & unstructured Data

- Unstructured data : images, language etc. (humans)
 - Structured data: transactions at a store etc. (Not human)
 - Current performance (algorithms) ?
 - Implementation
 - Helps establish error (irreducible etc.)
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ML Project - Getting started

- Iterative model building
 - Review current literature (current implementations)
 - Any open-source implementation would help set a baseline
 - Consider the deployment scenario and computational resources.
 - Overfit a small training set before using a larger dataset.
 - For example, image classification, train on 100 images.
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ML Project - Error Analysis

- Iterative model building
 - Review current literature (current implementations)
 - Any open-source implementation would help set a baseline
 - Consider the deployment scenario and computational resources.
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